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Development and Field Validation of an Automated Real-Time Drilling Fluid Measurement System

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Abstract

Traditional methods of rheological and density measurements using offline devices require manual intervention and are prone to human error. Additionally, decision-making relies on fluid property data that may be several hours old. Automating the drilling fluid measurement process enables drilling engineers to access real-time, reliable fluid data when making critical decisions. This paper presents recent advancements in the development of a real-time drilling fluid measurement unit, which minimizes safety risks for operators while providing accurate, consistent, and automated measurements of drilling fluid properties.

The unit continuously measures drilling fluid density, rheological parameters, temperature, and pH at a frequency suitable for critical decision-making. The compact system (1 m × 0.7 m × 0.9 m) consists of a circulation system, a measurement system, and a control system that supports both automatic and manual operation. A self-cleaning module ensures proper cleaning between cycles and allows quick switching between oil-based and water-based fluids. Fluid density is measured using an insertion type sensor on a static column, while rheology is determined using an automated Couette viscometer with variable speeds. A prototype was first built and tested in the laboratory before being deployed to an offshore well.

During field deployment, the unit collected and monitored parameters for several days using oil-based drilling fluid, and the results were compared with traditional offline instruments. The automated measurements showed close agreement with offline methods. Since the initial deployment, multiple units have been deployed, demonstrating reliable continuous operation with minimal error. The unit's outputs can be integrated into an in-house drilling hydraulic simulator to predict circulating pressures. A case study illustrates the application of the measured data alongside the hydraulic simulator to monitor equivalent circulating density (ECD) in a drilling well. Predicted ECD values were compared with pressure-while-drilling (PWD) measurements over a 1600-meter interval. The results show that the predicted ECD closely follows the PWD trendline, with an average error of about 1%.

This paper presents the results and lessons learned during the development of an in-line mud measurement unit, which enables continuous and accurate monitoring of fluid properties. It also provides a comparative analysis of in-line and offline measurements. The integration of the unit's measurements with a hydraulic simulator yields promising results for real-time ECD tracking during drilling operations.

Keywords: Real-time data, inline measurement, drilling fluid, rheology, ECD

Introduction

Downhole pressure control requires accurate knowledge of drilling fluid properties, including density and rheology, which are both temperature and pressure dependent (McMordie et al. 1982; Politte 1985; Zamora et al. 2013). It has been shown that ignoring such dependencies can significantly affect prediction on the equivalent static and circulating density of drilling fluids (Zamora et al. 2013; Dokhani et al. 2016). Thus, drilling fluid measurements are routinely conducted to monitor and optimize fluid properties, ultimately improving drilling performance. Nevertheless, these measurements are typically performed manually four times per day with equipment that may not be properly calibrated. Operators often make critical downhole decisions based on fluid properties that are 6 to 24 hours old, which may not accurately represent the current situation in the wellbore. In critical situations, there is no time to conduct tests for better recommendations. Consequently, decisions based on outdated fluid data can exacerbate hole problems. Quality decisions require consistent, accurate, and complete data.

Moving towards a digital oilfield has been part of the strategic plan for major oil and gas service companies. The need for automated real-time fluid measurements has long been recognized. The drilling industry envisions process automation to reduce human exposure to significant health, safety, and environmental (HSE) risks, as well as to improve operational efficiency and consistency in challenging drilling conditions. However, real-time monitoring of drilling fluid properties is associated with operational difficulties, including device specification limits, the presence of solids, and the need for routine cleaning.

Saasen et al. (2008) described the design of a drilling fluid analysis system, which was jointly developed by several major operating and service companies. However, the system was designed for oil-based muds. The authors reported results of yard tests, but no report of field deployment of an integrated system is given.

Broussard et al. (2010) explain the development of a unit to measure the real-time density and viscosity of drilling fluids. The unit includes an inline density meter that operates based on the Coriolis principle. The density meter also records temperature to be used as a reference for the measured density and rheology. The viscometer is a Couette (concentric cylinder type) with variable speed control. The size of the unit package is 1.2 m (H) $\times$ 1.2 m (W) $\times$ 0.6 m (D). The results of the density measurement indicate that the average discrepancy between the densitometer and mud balance is within 0.1 ppg. A comparison of the sensor measurement with the measurement of FANN 35 for high rpm and low rpm is shown. The authors also present rheology measurement results for water-based mud and conclude that the discrepancy with the traditional measurement method is due to the presence of solids. The lessons learned from the field trials were implemented in post-trial activities, including additional automated cleaning routines, re-routing of components of the internal cleaning system, and the addition of strainers to the suction line. An improved version, termed a density rheology unit (DRU), is presented by Dotson et al. (2017).

Carlsen et al. (2012) utilized instrumented standpipe to measure frictional pressure losses. The authors argued that measuring the drilling fluid rheology with a viscometer and then calculating frictional pressure losses can be replaced with a direct method of frictional pressure loss measurement across the standpipe. The method still requires constructing the flow curve (based on measure pressure losses at different flow rates) to obtain predictions for different flow condition.

Magalhães Filho et al. (2017) developed a flow loop to evaluate the performance of two inline viscometers: the TT-100 viscometer by Brookfield and a pipe rheometer, both of which run in parallel lines. Nevertheless, it is noted that the measurement range of the pipe rheometer is constrained at the lowest limit by the minimum speed of the pump and at the highest limit by the maximum flow rate required to maintain a laminar flow regime. The results of the inline viscometers show a slight divergence for shear rates above 750 s^{-1} .

Contreras et al. (2019) revealed a semi-automatic device for the measurement of drilling fluid rheology and density. However, the fluid sample needs to be poured into a container and manually attached to the device. Thus, the design is more favorable for laboratory work and is not considered suitable for continuous and real-time measurement.

Gul et al. (2020) developed an automated skid-mounted mud unit (MSU) for the measurement of density and rheological properties of drilling fluids. The unit is deployed in the Permian Basin. However, the system is not integrated with a software package for real-time prediction of downhole conditions.

Ombe et al. (2020) presented a Density Rheology Monitoring System (DRMS), which consists of the Density Rheology Unit (previously developed by Halliburton) that is linked to a Monitoring System. The role of the Monitoring System is to present the measured data and plot the trends to facilitate early detection of changes. However, the system does not integrate with software to simulate the downhole conditions.

In recent years, there have also been attempts to develop automated systems for the measurement of drilling fluid properties (e.g., Zhang et al., 2023, and Xin et al., 2025). The system developed by Zhang et al. (2023) employs a pipe viscometer for rheology measurement, a Coriolis flow meter for density measurement, and ion-selective electrodes for ionic concentrations. Although the developed system is designed in four configurations, the footprint of the displayed land unit is large (about 1.2 m × 2.4 m × 1.8 m), which can be a concern in offshore drilling rigs. Xin et al. (2025) presented a compact unit that is designed to monitor inlet and outlet drilling fluid properties. Although multiple modular designs are shown, the footprints of the individual and integrated systems are not specified. The authors did not detail the concept of rheological and density measurement; however, it is expected that the rheological properties were measured using a Couette viscometer. Although the rheological properties for a Herschel-Bulkley fluid are determined for a sample test fluid, it is questionable why the flow behavior index for multiple samples is reported to be 0.9900. In the case of two previous studies, these systems are not yet integrated with hydraulic software to evaluate downhole conditions.

The industry has recognized the need for automated real-time monitoring of drilling fluid properties to improve efficiency and safety, though challenges like device limitations, consistency and solid handling persist. Several systems have been developed, such as the Density Rheology Unit (DRU) and automated skid-mounted units, but many lack integration with predictive drilling software for downhole conditions. Recent advancements, like those by Zhang et al. (2023) and Xin et al. (2025), focus on modular designs for real-time monitoring, though issues like large footprints and incomplete integration with hydraulic software remain unresolved. Meanwhile, most of these systems focus on displaying the measured data and lack anomaly detection algorithms for drilling fluid properties. Overall, while progress has been made, fully automated, real-time fluid measurement systems with predictive capabilities are still evolving. This paper presents a recent development of a fully automated real-time mud measurement unit. An example of integrating measured data with previously developed drilling hydraulic software is also presented in the last section.

Design Features

The mud measurement unit is a fully automated unit that can measure the density, rheology, temperature, and pH value of drilling fluid. The device is designed with an explosion-proof structure and is waterproof and corrosion-resistant. It can be used on offshore drilling platforms and has various qualifications for offshore operations.

The data acquisition system is used to collect and store measurement data, realize data visualization, and thus enable real-time monitoring of the drilling fluid system. Users can control and monitor the operation of the mud measurement unit and obtain measurement parameters through the external LCD display screen. At the same time, the mud measurement unit has a self-cleaning function, and the overall footprint is less than 2 m × 1 m × 1 m. Connections for the drilling mud supply, and the washing fluid supply are

provided within the unit. The unit can handle both water-based and oil-based muds and does not interfere with the existing circulation system on the rig. The drilling mud density is measured in real-time, and full rheological measurements are performed every 25 minutes (at 6 regular speeds). It is worth noting that the rheological readings can be converted and reported as Herschel-Bulkley parameters (such as n , k , and τ_y). Measurements can be performed at other speeds if specified. Other parameters, such as pH and temperature, are measured and reported in real-time. The measurement data are collected and stored using a data acquisition system, which allows for data visualization and therefore enables real-time monitoring of the drilling fluid properties.

Subsystem Functions

The overall design is divided into three subsystems: the circulation system, the measurement system, and the control system. The circulation system includes a marine centrifugal pump to draw the drilling fluid into the measuring unit. In addition, the circulation pipeline also includes water pipes, several solenoid valves, and various accessories. The piping system is made entirely of 316 stainless steels. The design assumes that the drilling fluid level in the mud tank is above the facility's suction line and that the feed fluid is free of large solid particles. To achieve this, a filter can be installed upstream of the suction line to remove large solid particles from the fluid stream.

The measurement system includes a pH meter, a density meter, a temperature sensor, and an automated inline rheometer. The rheometer, pH meter, density meter and thermometer are all explosion-proof. The core component of the measurement system can be considered as the rheometer device for continuous viscosity measurement. The inline rheometer can operate at 6 different speeds (3rpm, 6rpm, 100rpm, 200rpm, 300rpm, 600rpm), which can meet the measurement requirements of water-based and oil-based drilling fluids. The inline rheometer produces a signal proportional to the viscosity of the process fluid. The measured angular displacement is expressed in terms of dial reading to be comparable to the offline viscometers. An insertion-type tuning fork density meter is installed on vertical pipe to measure the fluid density. We designed a self-cleaning function for the rheometer. An air pump is used to clean the residual liquid in the rheometer between each test to increase the accuracy of the test. An explosion proof glass electrode pH meter is selected which is ideal for industrial application and provide real-time monitoring of pH values.

The control module includes an AC power supply, a data acquisition system to control the operation of valves, pumps, and other measurement modules, and an LCD display panel for the operator to interact with the equipment, and software system to interact with the user. The data acquisition system includes an interface to display and track the measured data as well as a status control interface. The system design follows plug-and-play functionality for seamless integration with the mud system. The interface follows intuitive operation with minimal training. The system can store measured data for historical data logging.

The unit is equipped with eye bolts for easy lifting. The system can operate with an ambient temperature up to 50°C. [Fig. 1](#) shows the front and back side of the mud measurement unit. A list of major measurement parameters of the unit is given in [Table 1](#).



Figure 1—Mud Measurement Unit, front side (left), back side (right)

Table 1—Operating range of mud measurement unit

Parameter	Range	Error
Density, g/cm ³	0-2.5	±0.01
Rotational viscometer, dial reading	0-330	±1
pH, dimensionless	0-14	±2% FS
Fluid temperature, °C	0-85	±0.5

Prototype testing

Several tests were conducted to compare the performance of the prototype rheometer versus the reference rheometer. Here, we show a comparison of the two viscometers in Fig. 2 when different solutions of Xanthan Gum solutions were used. The figure shows that the measured data are almost consistent between the prototype and the reference, and all deviations are within ±1 in terms of dial reading.

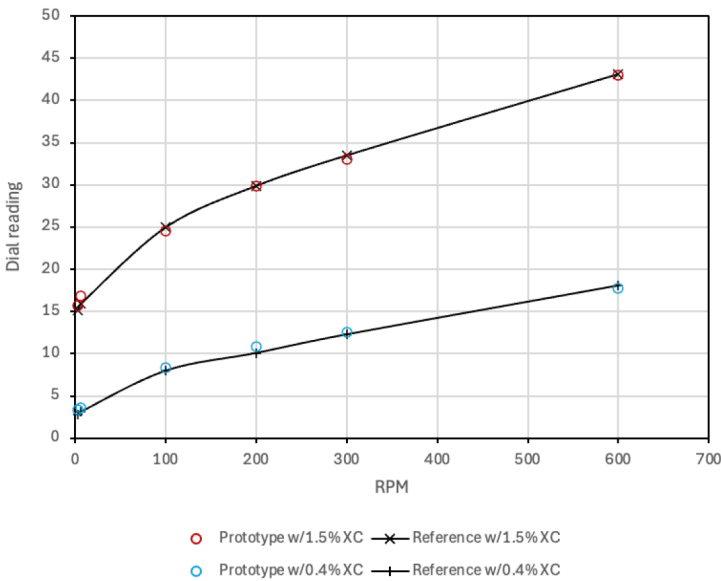


Figure 2—Comparison of the measured data of prototype unit versus reference viscometer for two different XC solutions

Another set of tests was conducted using two different densities of oil-based muds. The original mud prepared at the field (referred to as high-density mud) has more solid content, hence a higher density and higher viscosity. One barrel of the drilling mud was also centrifuged to separate part of the solid phase and is referred to as low-density mud. The test started at 12:00 on August 29th and ended at 15:30 on September 1st. It was conducted in a 24-hour continuous measurement mode. During the test, the data were recorded every 2 hours, and samples were taken for manual testing to be compared with the reference viscometer, densitometer and thermometer. The early tests were conducted with low-density mud (centrifuged mud). Then, the test fluid was replaced with a high-density mud, and the tests were continued for two days. Finally, the test fluid is replaced with a low-density mud. Fig. 3 displays the measured density and temperature by the unit versus those of the reference instruments. The temperature error is basically controlled within $\pm 0.5^{\circ}\text{C}$; and the large difference in density measurement is perhaps caused by inaccurate volume control of manual sampling.

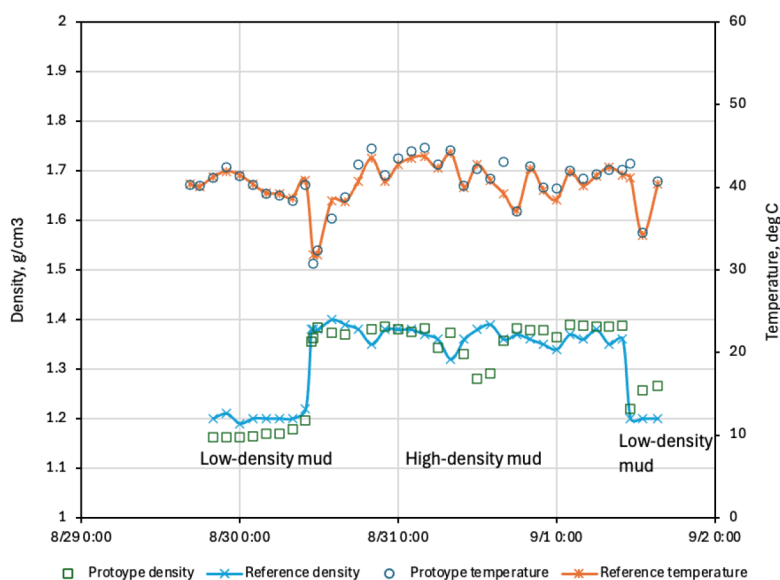


Figure 3—Comparison of the measured density and temperature prototype unit versus reference instruments for different oil-based muds

A comparison of the measured data at 600 rpm and 3 rpm by the prototype unit and a reference viscometer is displayed in Fig. 4. The average discrepancy between the prototype unit and the reference viscometer is less than 0.2 in terms of dial reading. It shall be noted that the rheological outputs of the prototype unit are expressed in terms of dial reading to be comparable with an offline viscometer. During the replacement of the mud liquid, the detection recording frequency was appropriately increased. The test results show that the measurement accuracy of the prototype for various parameters of drilling fluid is acceptable. The error of viscosity at six speeds is controlled within ± 1 . The results further show that the error value of viscosity measurement has no obvious temperature characteristics. The prototype can work continuously and maintain stable measurement accuracy during continuous operation. After replacing the mud, the measurement accuracy basically returns to normal with proper conditioning and tends to yield stable results after 30 minutes.

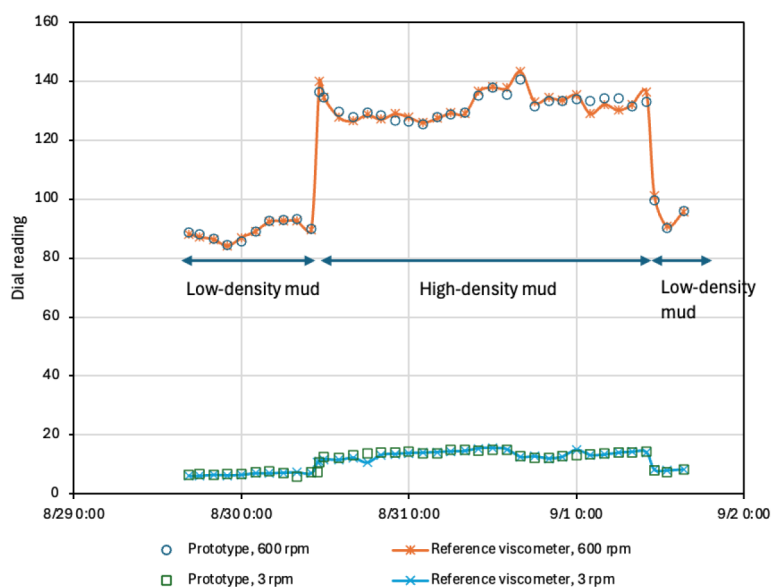


Figure 4—Comparison of the measured data of prototype unit versus reference viscometer for different oil-based muds

Field Study

After receiving all the necessary certificates, the developed unit could then be deployed for the first trial. The equipment ran continuously for three days on an offshore rig A in East Asia. The unit was shut down after running overnight to check particle screening filters. For the next two days, the unit remained operational with no problems. After three days, when the platform was powered off, the equipment was shut down and data recording was stopped. The unit was shut down for inspection a few times. A comparison of viscosity, density, and pH measurements from the mud measurement unit versus manual measurement devices is shown in Fig. 5. The plot shows that the unit was constantly in service for over 24 hours with no problems. Comparing the measured values of the unit versus manual measured data shows satisfactory agreement, although these measurement devices have different accuracies. Due to personnel limitations, it is difficult to keep pace with the changes in drilling fluid parameters. However, an automated system can quickly detect the changes, as displayed by an upward trend in rheology and density of the fluid towards the end of monitoring period in Fig. 5.

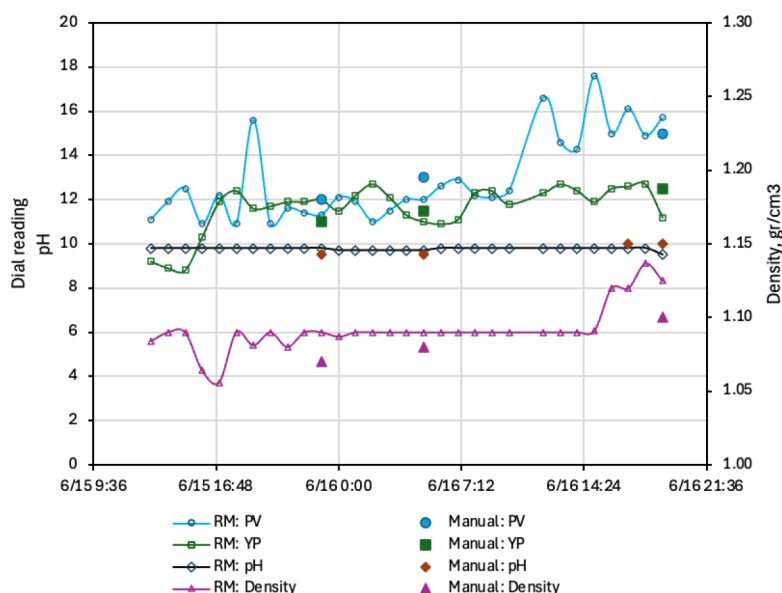


Figure 5—Comparison of viscosity, density and pH measurement from the unit versus manual measurement in rig A

The next two trials were also conducted in an offshore field in East Asia. The depth of water exceeds 300 m. The unit was used while drilling a 12-1/4" section for the depth interval of 1270 m to 3340 m. In both trials, the unit was installed near the mud tank and operated for a few days with water-based mud (Fig. 6). An interface of the main display of the automated mud measurement system is shown in Fig. 7. Like the previous case, the measured data were then validated with offline measurement devices. A comparison of the measured rheological parameters and density of the unit versus those of manual devices for well E19 is depicted in Fig. 8. The plot indicates that the performance of the unit is acceptable when compared with manual reference measurements, although the manual measurements were not conducted every hour. In well E20, the comparison was conducted for half a day, and the results are shown in Fig. 9. The automated unit shows that the measured parameters are experiencing some fluctuations, but these are well controlled during the given period.



Figure 6—Mud measurement unit in operation near the mud tank on an offshore drilling platform

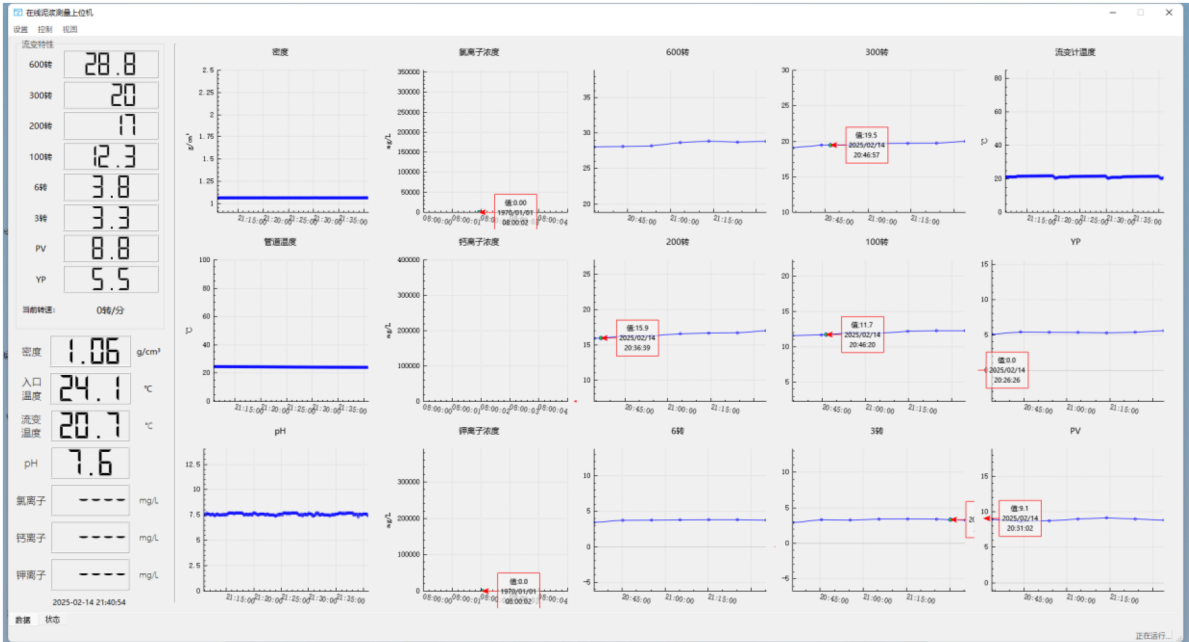


Figure 7—Interface of automated mud measurement unit

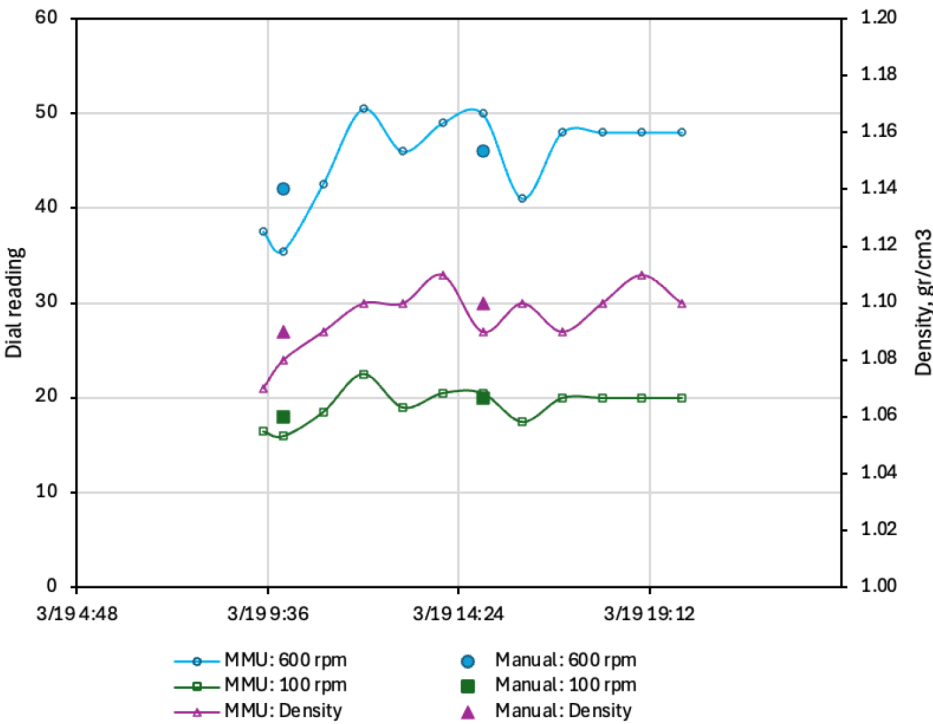


Figure 8—Comparison of viscosity and density measurement from the unit versus manual measurement in Well E19

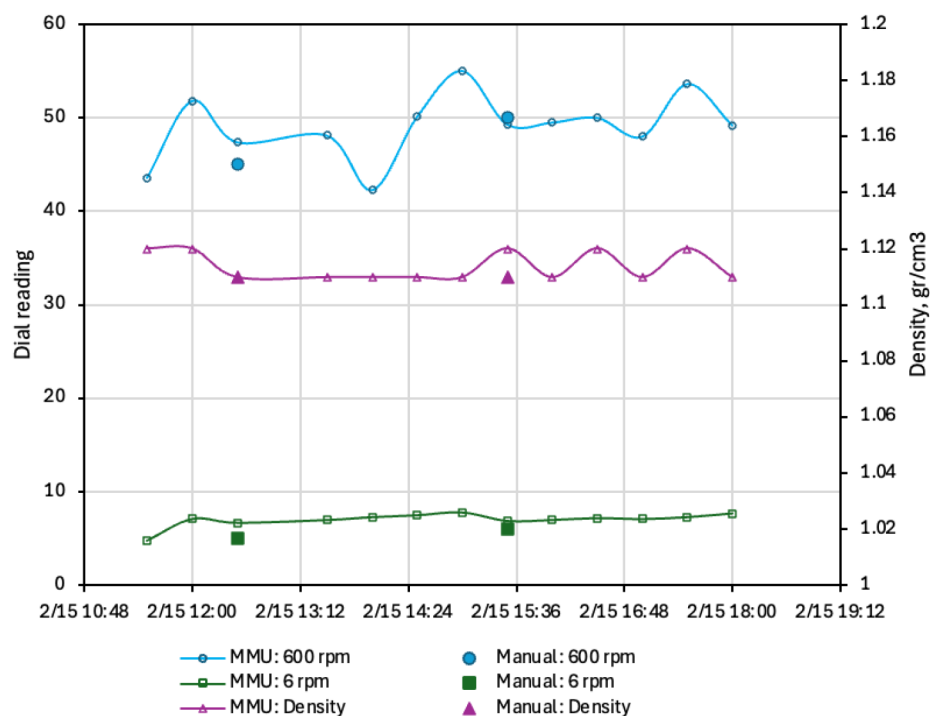


Figure 9—Comparison of viscosity and density measurement from the unit versus manual measurement in Well E20

The last field trial was conducted in the 8-1/2" section of well H2. The tests were conducted for 3 days, during which water-based mud was used. A comparison of the measured results of the unit and those reference values is depicted in Fig. 10. The measured density values are consistent with the manual measurement. The measured values of R600 and R6 are also within ± 1 of the manual viscometer. The graphs further indicate that some measurement values (particularly R600) of the unit can fluctuate; however, the user needs to consider averaging the past few measurements or using a trend line rather than focusing on a single measured data point.

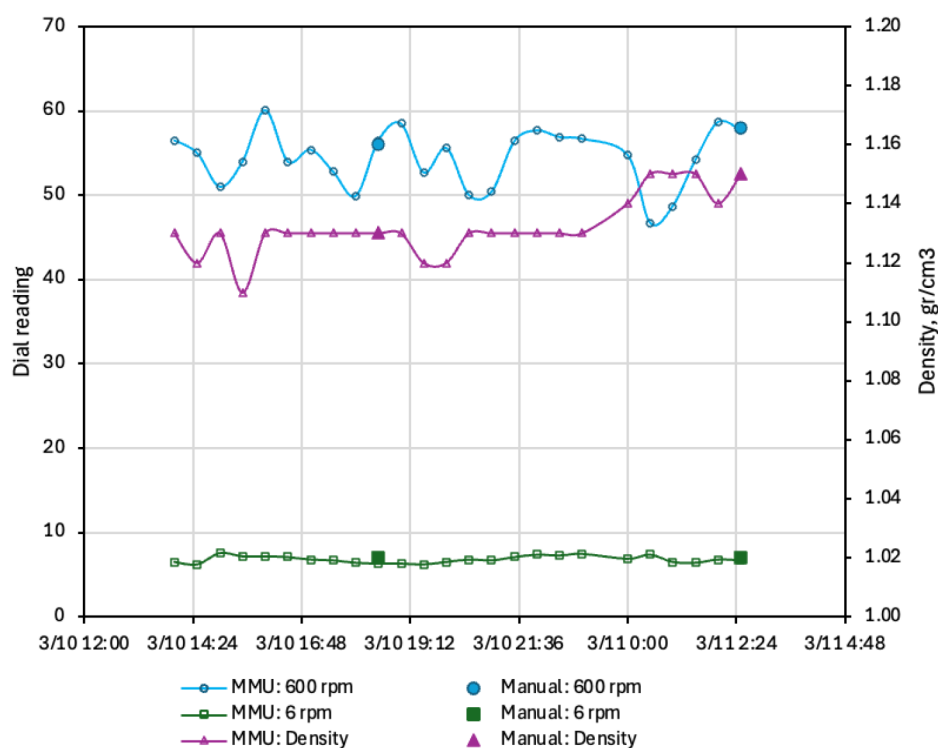


Figure 10—Comparison of viscosity and density measurement from the unit versus manual data in Well H2

It is worth noting that the mud measurement unit measures the density and rheological parameters of the fluid based on the in-situ temperature of the mud tank. However, the manual measurements are conducted on a limited sample volume, which is susceptible to cooling effects due to the time lapse between the sampling and the measurement by mud engineer.

The field tests reveal a necessity for improvement in a few areas. For example, it has been realized that employing external filters to prevent abnormal objects from entering the equipment in extreme cases may be necessary, as this could cause fatal damage to the instruments. An improved design has been made on the external filters to extend the time between regular cleanings. An upgrade was also made to the circulation pump to ensure that the lubricating oil level is always sufficient to avoid loss of lubricant and consequently failure.

Real-time Monitoring

Based on the communication protocol of the Wellsite Information Transfer Standard (WITS), the data is imported by proprietary real-time drilling hydraulic software. The software also obtains real-time data on drilling fluid properties from the automated mud measurement unit. The data transmitted via WITS or WITSML (Wellsite information transfer standard markup language) protocols is used by proprietary real-time drilling hydraulics software to simulate downhole pressure, temperature and cutting transport. The drilling hydraulic software can operate in two modes. The first is a real-time mode, where the input data from the rig and mud measurement unit can be imported into the software. Then, the software interactively runs with every feed of entry data and returns the predicted ECD values. The second mode is a batch simulation, which is more suitable for design purposes. In this mode, the sequence of events must be specified by the operator. Each event is a set of drilling parameters for a given period, which includes bit depth, circulation rate, rate of penetration, weight on bit, fluid density, PV, YP, etc. The table of events is then run sequentially, and the outputs are calculated based on the specified time frame for the output parameters.

The application of an integrated system of the mud measurement unit and hydraulic modeling is investigated in an offshore well for an interval where the data for pressure while drilling (PWD) was

available. The depth of water is less than 100 m. The last casing is 13-3/8" and casing shoe depth is 1800 m. The borehole diameter is 12-1/4", drill pipe outer diameter is 5-7/8" and drill collar outer diameter is 8". The wellbore has an inclined trajectory with a maximum inclination angle of 80° and a slant section of about 3800 m. The selected drilling interval is from 1800 to 3600 m, all in one section. The measured ECD data (from the PWD tool) and predicted ECD are compared in Fig. 11. At the start of 1800 m, the ECD was about 1.07 g/cm³, and at the depth of 3600 m, the ECD value reached 1.22 g/cm³. The pattern shows three regions in terms of ECD versus depth. In the first region (1800-2350m) the ECD increases from 1.07 to 1.17 g/cm³. In the second region (2350-3150 m), the ECD slightly increases from 1.17 to 1.2 g/cm³. In the last section, the ECD increases from 1.2 to 1.22 g/cm³. The comparison between model prediction and measured PWD data shows that the model can capture the trend with reasonable accuracy. The average and maximum discrepancy between the measured PWD data and the model predictions are about 0.003 g/cm³ (0.027 lbm/gal) and about 0.014 g/cm³ (0.11 lbm/gal), respectively. However, the specific events that caused a spike in PWD data are unavailable to us. Those events could include surge/swab phenomena, which are not defined in the current model package. While the current hydraulic simulation package does not include surge/swab predictions, our prior publications demonstrate how to integrate them for wellbore stability analysis (Liu et al. 2023, Ma et al. 2024). It is shown that the computation time for surge/swab pressure calculation is more intensive than for transient hydraulics because the former requires integration with the transient effects of pipe movement. Meanwhile, the velocity profile and the speed of tripping pipe, which highly affect the magnitude of surge/swab pressure, are typically not available during tripping in field operations. Although the model was later run in the office after acquiring all the necessary inputs, the presented example demonstrates the value of an integrated system of real-time measurement and monitoring.

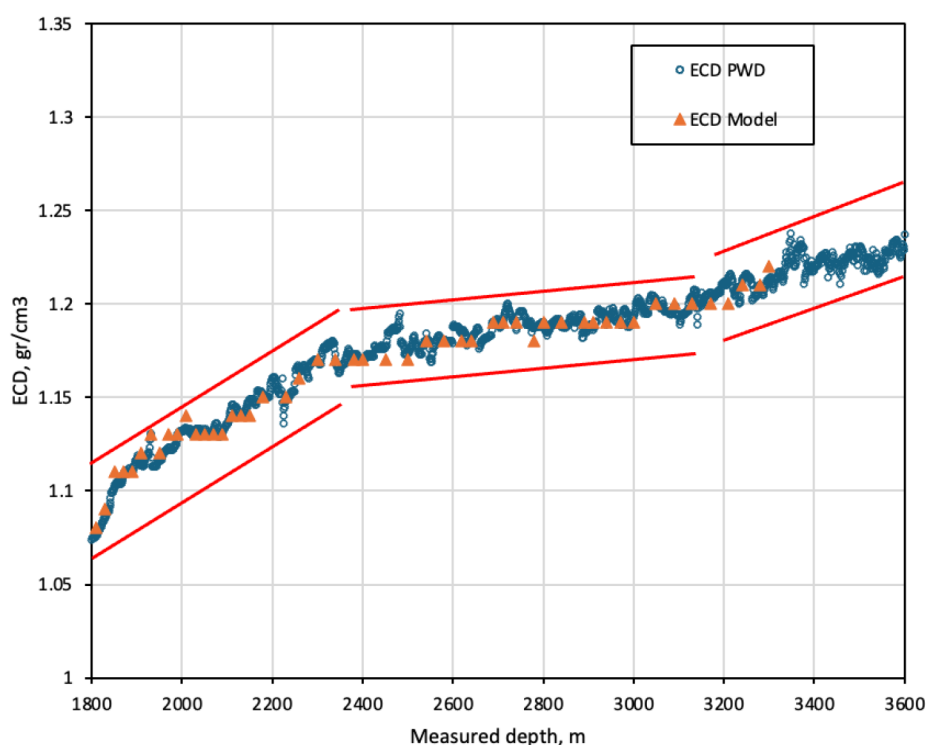


Figure 11—Comparison of the measured downhole ECD data and predicted ECD of the model

Conclusions

The development and field deployment of an automated, real-time mud measurement unit demonstrate significant advancements in improving the accuracy, consistency, and safety of drilling fluid monitoring. The designed unit provides continuous measurements of drilling fluid density, rheological properties,

temperature, and pH with minimal error compared to traditional offline methods. Laboratory tests confirm that the automated measurements closely match manual reference data, ensuring reliable decision-making based on real-time fluid properties. The system's compact footprint and explosion-proof, corrosion-resistant construction make it suitable for onshore and offshore drilling environments. The self-cleaning module enhances operational efficiency by allowing quick switching between oil-based and water-based fluids while maintaining measurement accuracy. The presented field trials in offshore wells validated the unit's performance under drilling conditions. The system operated continuously with minimal downtime, demonstrating its ability to detect fluid property changes more efficiently than manual sampling methods.

By feeding real-time drilling fluid data and rig data into a proprietary hydraulic simulator, the system enables accurate prediction of equivalent circulating density (ECD). A case study comparison with pressure-while-drilling (PWD) measurements showed an average error of about 1%, confirming the model's reliability for real-time wellbore pressure management.

Automation reduces human intervention, minimizing exposure to hazardous conditions and eliminating inconsistencies from manual sampling. The system's ability to detect fluid property changes in real time allows for proactive adjustments, mitigating drilling risks such as lost circulation or wellbore instability. It is recommended to enhance data analytics for anomaly detection in fluid properties and incorporate surge/swab modeling for tripping operation to improve ECD predictions.

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